Category: Cloud Computing & IoT

PS-033: DNS-Based Discovery of Diagnostic Services

- **Description:**
  - Diagnostic capabilities in distributed networks, such as latency probes and traceroute responders, are often configured using hardcoded IP addresses or proprietary registries that break when endpoints move. The IETF DNS-AID proposal explores using the Domain Name System to allow clients and automated agents to discover service endpoints dynamically. In this project, based on draft-mozleywilliams-dnsop-dnsaid, you will configure a DNSSEC-signed test zone publishing structured metadata for three mock diagnostic services. You will build a client that discovers endpoints via DNS queries, validates record authenticity using DNSSEC, and performs an authenticated invocation, demonstrating proper error handling for bogus DNS data and unavailable services.
- **Challenge:**
  - Prototype discovery of three lab diagnostic capabilities using DNS-AID-style metadata, DNSSEC validation and an authenticated invocation step. The system must distinguish valid metadata from bogus or malformed DNS responses.
- **Key Requirements**
  - Use 1 signed DNS test zone with DNSSEC enabled.
  - Implement 3 mock diagnostic services.
  - Use 1 discovery client.
  - Follow a pinned subset of the DNS-AID draft.
  - Implement DNS-based service metadata discovery.
  - Validate discovered records using DNSSEC.
  - Use an authenticated invocation step after discovery.
  - Test valid discovery, bogus DNS data, malformed metadata and unavailable services.
  - Clearly separate metadata integrity from service authorisation and trust.
  - No claim that DNS discovery authorises invocation or proves service trustworthiness.
  - No Internet-wide service registry.
- **Expected Outcome**
  - DNS-AID-style metadata records
  - Publisher component
  - Discovery client
  - Three mock diagnostic services
  - DNSSEC validation mechanism
  - Authenticated invocation workflow
  - Validation and negative test cases
  - Handling for malformed, bogus and unavailable services
  - Technical report and demonstration
- **Deliverables***

1. GitHub Collaboration: (**This step is crucial for allowing mentors to provide real-time oversight.**)

- Each team must set up a private repository on GitHub.
- Add our GitHub account (https://github.com/aiori-hackathon) as a collaborator.
- Once the collaboration request has been accepted all the concerned users (all the teammates as well as aiori-hackathon account) transfer the ownership of the repository to our GitHub account. (https://github.com/aiori-hackathon)
- Ensure that your repository is named according to your team's name.

2. Each team is required to upload a PDF to their GitHub repository
   - Please follow the specified structure: Proposed-structure-hackathon.pdf
3. Prototype Demonstration with a Pseudocode Snippet to the mentor

- **References**
  - IETF Draft | DNS-AID: AI Agent Discovery via DNS
  - IETF 126 Hackathon